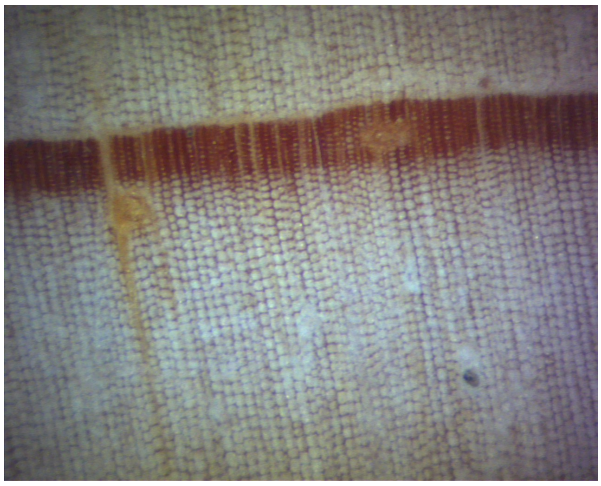
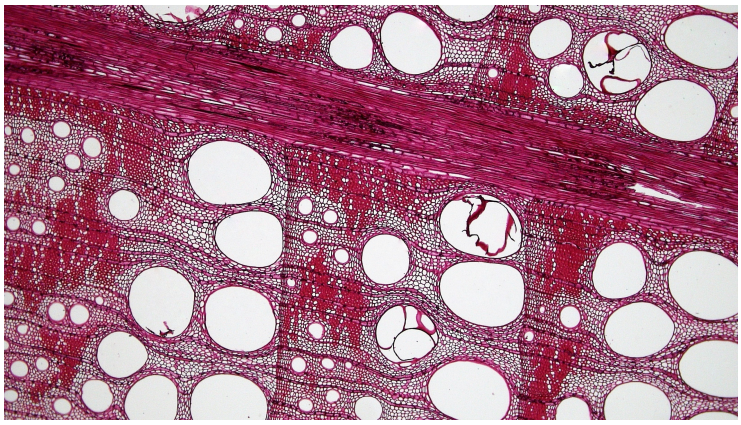
Lesson 2 Compare real wood anatomy

For slides 15–17. These are different specimens at different magnifications; do not compare cell sizes directly between panels.



Scots pine (Pinus sylvestris). Full image width approximately 2.1 mm. Examine changes in tracheid lumina and walls.

Taxelson, 2009. Public domain.

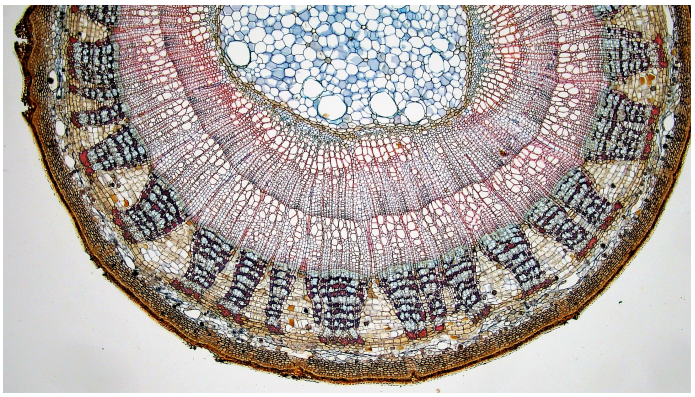


Quercus wood. Compare the band of large earlywood vessels with the tissue beyond it. Original label: 40x; physical dimensions on this page are not calibrated.

Berkshire Community College Bioscience Image Library. CC0 1.0.

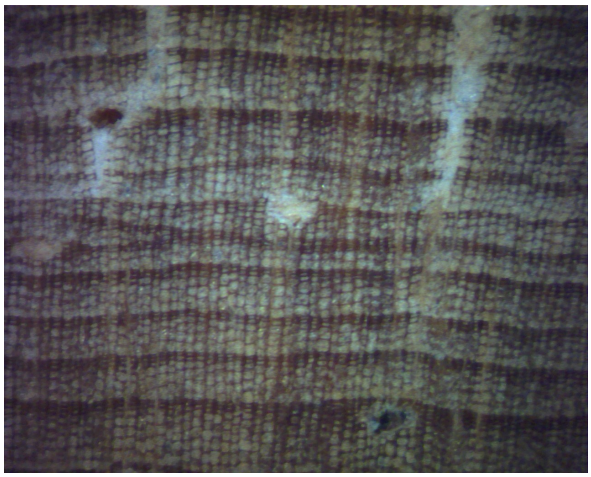
Locate a candidate ring boundary in each image and describe one feature other than colour that supports it.

Lesson 2 Diffuse porosity and discontinuous rings



Young Tilia stem. Examine secondary xylem around the pith. Are large vessels concentrated at the start as strongly as in ring-porous wood? Physical scale is not calibrated.

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Scots pine. Full image width approximately 2.1 mm. Follow a narrow ring across the image.

Taxelson, 2009. Public domain.

If a ring cannot be seen in one direction, can we conclude that the whole tree stopped growing that year?

Interpretation

Tracheid lumina and walls help identify conifer ring boundaries. Concentrated large earlywood vessels help in ring-porous wood. In diffuse-porous wood, also examine boundary cells and their arrangement. A discontinuous ring pinches out locally: check other radii and specimens. This image alone cannot establish the number of absent years.

Staining and lighting affect colour. A false ring is an intra-annual feature, distinct from a locally discontinuous ring. The existing reference link to Japanese birch (Betula maximowicziana) remains available.